

Project Report: Income Prediction Classification Model

1. Executive Summary

This report details the development of a predictive classification model designed to identify high-income individuals (earning $\$ > 50\text{K}$) for a retail client. By utilizing a **Random Forest** architecture, the model achieved a high accuracy of **93%**. This model serves as a lead-qualification tool, allowing the business to focus marketing resources on individuals with verified high-value potential.

2. Data Exploration & Pre-processing

Exploration Strategy

- The primary objective was to determine which demographic and behavioral features correlate most strongly with wealth.
- Initial exploration revealed that **Age** and **Weeks Worked in Year** were the top two drivers of income levels.
- We observed that **Capital Gains** were highly skewed; while most individuals had zero gains, those who did possess them were significantly more likely to be in the high-income bracket.

Pre-processing Approaches

- **Feature Selection:** We prioritized 7 key columns that represent customer behavior: age, education, marital stat, major occupation code, sex, weeks worked in year, and capital gains.
- **Normalization:** Numeric features were processed using StandardScaler. This was a critical **business judgment** to ensure that large-scale numbers (like capital gains) did not drown out smaller-scale but equally important numbers (like age).
- **Label Encoding:** Categorical variables like "occupation" and "education" were converted into numerical formats to be readable by the Random Forest algorithm.

3. Model Architecture & Training

Model Selection

- **Algorithm: Random Forest Classifier.**
- **Reasoning:** We chose an ensemble method because retail data is often non-linear and contains outliers. Random Forest is robust against noise and provides an inherent "Feature Importance" ranking, which is valuable for explaining the model to business stakeholders.

Training Algorithm

- The data was split into training and testing sets to ensure the model could generalize to new customers.
- The model was trained to predict a binary output: **0** ($\leq 50\text{K}$) or **1** ($> 50\text{K}$).

4. Evaluation Procedure & Business Judgment

Evaluation Metrics

- The model reached an overall **Accuracy of 93%**.
- We utilized a **Confusion Matrix** to monitor False Positives (labeling a low-earner as high-income) and False Negatives.

Strategic Decision: Threshold Tuning

- **Business Logic:** In luxury retail, the cost of a "False Positive" (sending expensive marketing materials to someone who cannot afford the product) is high.
- **The Adjustment:** We moved the classification threshold to **0.7**.
- **Impact:** By requiring a 70% probability before labeling someone as "High Income," we ensured that the resulting target list is highly refined, maximizing the return on marketing investment (ROI).

5. Interesting Findings

- **Stability over Education:** Interestingly, "Weeks Worked" appeared more influential in predicting high income than "Education" alone. This suggests that "Experience" and "Persistence" are better predictors of current wealth than historical academic achievements in this specific dataset.
- **Segment Overlap:** High-income individuals were almost exclusively found in the "Executive" and "Professional Specialty" occupation codes.

6. Model Usage Recommendations

- **Tiered Marketing:** We recommend using the model to create a "VIP" list for anyone with a probability score > 0.85 for exclusive previews.
- **Lead Filtering:** For digital ad campaigns, the 0.7 threshold should be the default to maintain brand exclusivity and reduce ad-spend waste.
- **Continuous Learning:** The model should be re-trained every quarter as economic conditions and labor market trends shift.

7. References

- *Scikit-Learn Documentation:* Random Forest Classifier and Feature Importance implementation.
- *Pandas & Matplotlib:* Data manipulation and visualization of the confusion matrix.
- *StandardScaler Documentation:* Mathematical foundations for feature standardization.

Project Report: Customer Segmentation Model (K-Means)

1. Executive Summary

While the Classification Model identifies "who" has the financial capacity to purchase, this **Customer Segmentation Model** explains "how" to communicate with different subsets of the population. By utilizing **Unsupervised Learning (K-Means)**, we identified **5 distinct customer personas**. This allows the retail client to move away from a "one-size-fits-all" approach and instead deploy personalized marketing strategies that resonate with the specific lifestyle and career stage of each group.

2. Data Exploration & Pre-processing

Exploration Strategy

- The goal was to find natural groupings in the data that weren't immediately obvious.

- We explored multi-dimensional relationships between **Age, Work Intensity, and Investment Wealth**.
- **Interesting Finding:** We noticed that "Capital Gains" were not distributed normally; they were concentrated in a very small, high-value group, which heavily influenced the formation of our "Wealthy Specialist" cluster.

Pre-processing for Clustering

- **Feature Selection:** We selected 7 behavioral features: age, education, marital stat, major occupation code, sex, weeks worked in year, and capital gains.
- **Standardization (Critical Decision):** We used StandardScaler to normalize the data.
 - **Business Judgment:** K-Means relies on "Euclidean Distance." If we hadn't scaled the data, the model would have assumed that a \$1,000 difference in capital gains was 1,000 times more important than a 1-year difference in age. Scaling ensures every feature has an equal opportunity to define a cluster.

3. Model Architecture & Training

Algorithm Selection

- **Algorithm: K-Means Clustering.**
- **Architecture:** Centroid-based clustering that iteratively assigns points to the nearest cluster head.

Determining the Optimal Clusters (The Elbow Method)

- We ran the algorithm for a range of clusters (\$K=1\$ to \$10\$) and plotted the "Inertia" (Sum of Squared Distances).
- **Finding:** The "Elbow" clearly appeared at **\$K=5\$**.
- **Business Judgment:** While \$K=3\$ might have been simpler, it grouped "Executives" and "Retired Investors" together. By choosing \$K=5\$, we gained enough granularity to separate different types of high-value customers without making the marketing strategy too complex to manage.

4. Evaluation & Segment Profiling

Since clustering is unsupervised, evaluation was performed by "profiling" the segments against our known income labels.

Segment	Persona Name	Characteristics	Income Profile
Cluster 4	Executive Elite	Mid-age (46), Highest work weeks (48)	100% High Income
Cluster 2	Wealthy Specialists	Highest Capital Gains (\$99k average)	88.2% High Income
Cluster 3	Stable Workforce	Mid-age (38), High work weeks (48)	0% High Income
Cluster 1	Senior Demographics	High Age (62), Low work weeks	0% High Income
Cluster 0	Young/Entry-Level	Lowest Age (11), No work weeks	0% High Income

5. Business Judgment & Model Usage

Model Selection Rationale

We chose K-Means because it is computationally efficient and produces "circular" clusters that are easy to explain to non-technical stakeholders.

Usage Recommendations

1. Targeted Product Placement:

- Offer **Luxury Business Travel** and **Corporate Credit Cards** to **Cluster 4**.
- Offer **Investment Portfolios** and **Stock Market Tools** to **Cluster 2**.

2. Cost Savings:

- De-prioritize **Clusters 0 and 1** for high-cost mailers or premium sales calls, as their probability of high-income conversion is statistically zero.

3. Nurturing Strategy:

- Use **Cluster 3** (Stable Workforce) for "Growth" products, as they are economically active but not yet in the high-income bracket.

6. Suggestions & Future List

- **Refining with Geography:** Adding zip codes or regions could help the client identify *where* these clusters live for physical store placement.
- **A/B Testing:** Run a test where one group receives "Segment-Specific" ads and a control group receives "General" ads to measure the uplift in conversion.
- **Cluster Migration:** Monitor how customers move from Cluster 0 (Young) to Cluster 3 (Stable) to identify the "Golden Window" for acquiring lifelong customers.

7. References

- *Scikit-learn documentation* on K-Means and the Elbow Method.
- *Pandas* for data aggregation and segment profiling.
- *Matplotlib* for visualization of the Elbow Curve.